

SSTO fusion powered spaceplane using compact CHARM fusion reactor and 3-cycle electric rocket engine

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Abstract

We specify a single-stage-to-orbit (SSTO) spaceplane that flies Space Shuttle-style operations—including a Shuttle-class cargo bay—from a municipal airport to International Space Station (ISS) altitude in low Earth orbit (LEO), powered by a continuous Chambered Aneutronic Rotating Mirror (CHARM) $p\text{-}^{11}\text{B}$ plant [1] with direct energy conversion (DEC). Flight regimes use three electric stages: ducted fan on free air, microwave air plasma on climb, then carried-water plasma with intakes sealed. Each design step is written as a closed set of equations. We guesstimate the reactor mass hole, constrain water as a function of dry mass and vacuum Δv , impose a 1 GW plant with space restart and DEC, and solve a reference all-up mass. Combined-cycle engine maps and CHARM size/performance constraints follow.

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1 Vehicle vision

Municipal runway to ISS-class LEO: a Shuttle-style SSTO spaceplane with a real cargo bay, a single-deck crew module, and a continuous CHARM $p\text{-}^{11}\text{B}$ plant driving a three-stage combined-cycle engine (electric ducted fan $\rightarrow$ microwave air plasma $\rightarrow$ water plasma). The figures below are the vehicle picture; the equations that close the mass and energy budgets follow.

1.1 AI concept render

Fig.~1 is an AI-rendered concept illustration, not an engineering drawing: **Google Nano Banana Pro** (Gemini 3 Pro Image) was given the two CAD wireframes below as reference images and prompted to render the vehicle's true fuselage/wing contour — the grey wireframe lines, not the blocky bounding box — taking off from a municipal airport [21]. Proportions, panel lines, livery, and the runway scene are AI interpretation; the dimensioned CAD wireframes immediately below remain the source of truth for station layout and outer mold line.



Figure 1 CATSKILLS SSTO, AI concept render (Google Nano Banana Pro / Gemini 3 Pro Image) of the vehicle taking off from a municipal airport, prompted from the CAD floorplan and profile wireframes below.

Fig.~2 is a companion AI concept render of the flight deck at the same moment, from the rear seats looking forward: six crew in Crew-Dragon-style seats and suits (captain and copilot forward, four aft), helmets marked “Catskills Research,” small oval windows sized to match Fig.~1, and Crew-Dragon-style forward flight-deck controls. Illustrative only — seat count, suit design, and cockpit layout are AI interpretation prompted from the interior floorplan schematic (Fig.~3), not an engineering cabin design.



Figure 2 CATSKILLS SSTO crew cabin at liftoff, AI concept render (Google Nano Banana Pro / Gemini 3 Pro Image): view from the rear seats of the flight deck as the vehicle rotates off the runway, prompted from the interior floorplan schematic and the prior exterior-render window shape.

1.2 Interior floorplan and exterior profile

Crew volume flattens the **Space Shuttle crew module** from two decks to **one** [12,13], then **stretches** the pressurized nose so life support and a suited airlock are not cartoon-thin. Reference overall length $L \approx 52$ m. ECLSS = **Environmental Control and Life Support System**. Depth ≈ 6.5 –7 m, span ≈ 28 m.

Figs.~3 and 4 are orthographic CAD views of the same station map as `assembly.json` (nose left, length 52 m): crew capsule 0–11 m (flight deck + seats, internal O_2/N_2 , port **ground-only** side hatch); airlock 11–15 m (hatches cabin $\leftrightarrow$ airlock and airlock $\leftrightarrow$ bay only); cargo bay 15–33.3 m (18.3×4.6 m, clamshell doors on the cargo-skid drop-in; exterior OML is an unbroken tube); **fusion electric plant** 33.3–49 m on one skid (flight battery 33.3–35.5 m, water tanks 35.5–39.5 m — relocated ahead of CHARM as a supplemental radiation shield, §9.3 — fuel services 39.5–41.5 m, CHARM island incl. permanent shield bulkhead 41.5–49 m); **combined-cycle engine** 49–52 m (stages 1–3 + nozzle). The plant schematic (Fig.~9) is 1–1 with that JSON tree. Fig.~3 is a top-down cutaway (no landing gear). Fig.~4 shows white upper OML, dark TPS belly, extended gear, the port crew hatch, and closed top bay doors.

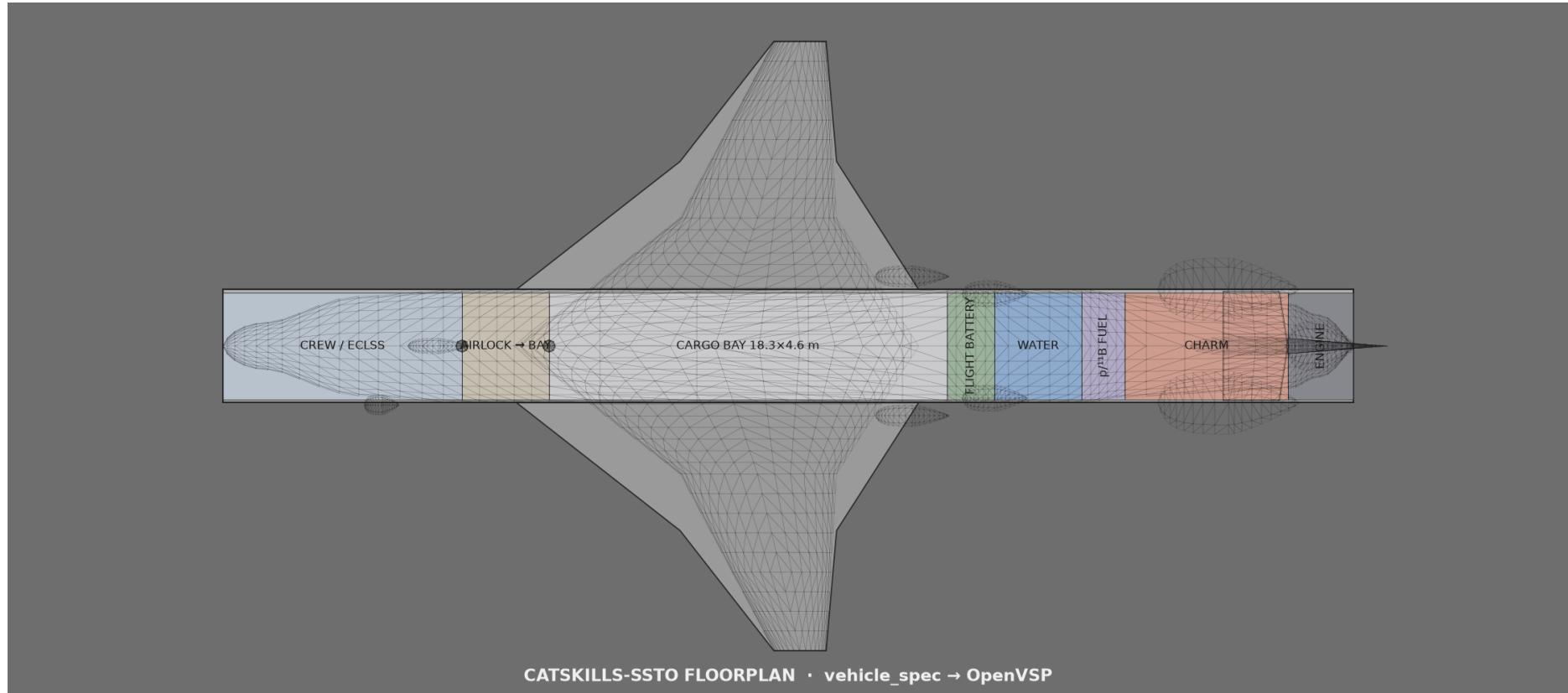


Figure 3 Vehicle floor plan.

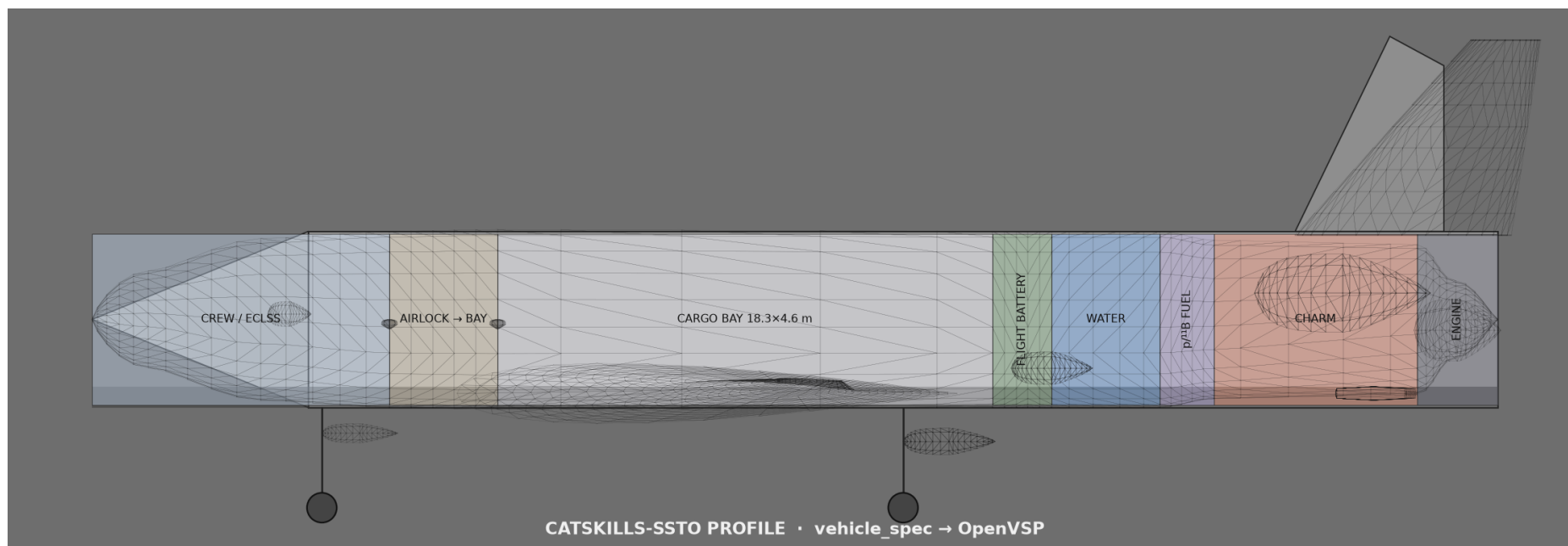


Figure 4 Vehicle profile view.

1.3 Forward drop-ins (top-down, covers off)

Figs.~5–8 are **Blender** orthographic top views built procedurally from `assembly.json` (make `cad-drop-ins`; see `research/figures/cad/build_crew_capsule_blender.py`, `build_airlock_blender.py`, `build_cargo_skid_blender.py`, `build_fusion_plant_skid_blender.py`, and the shared helpers in `research/figures/cad/lib/`). No AI-generated imagery remains in this section.

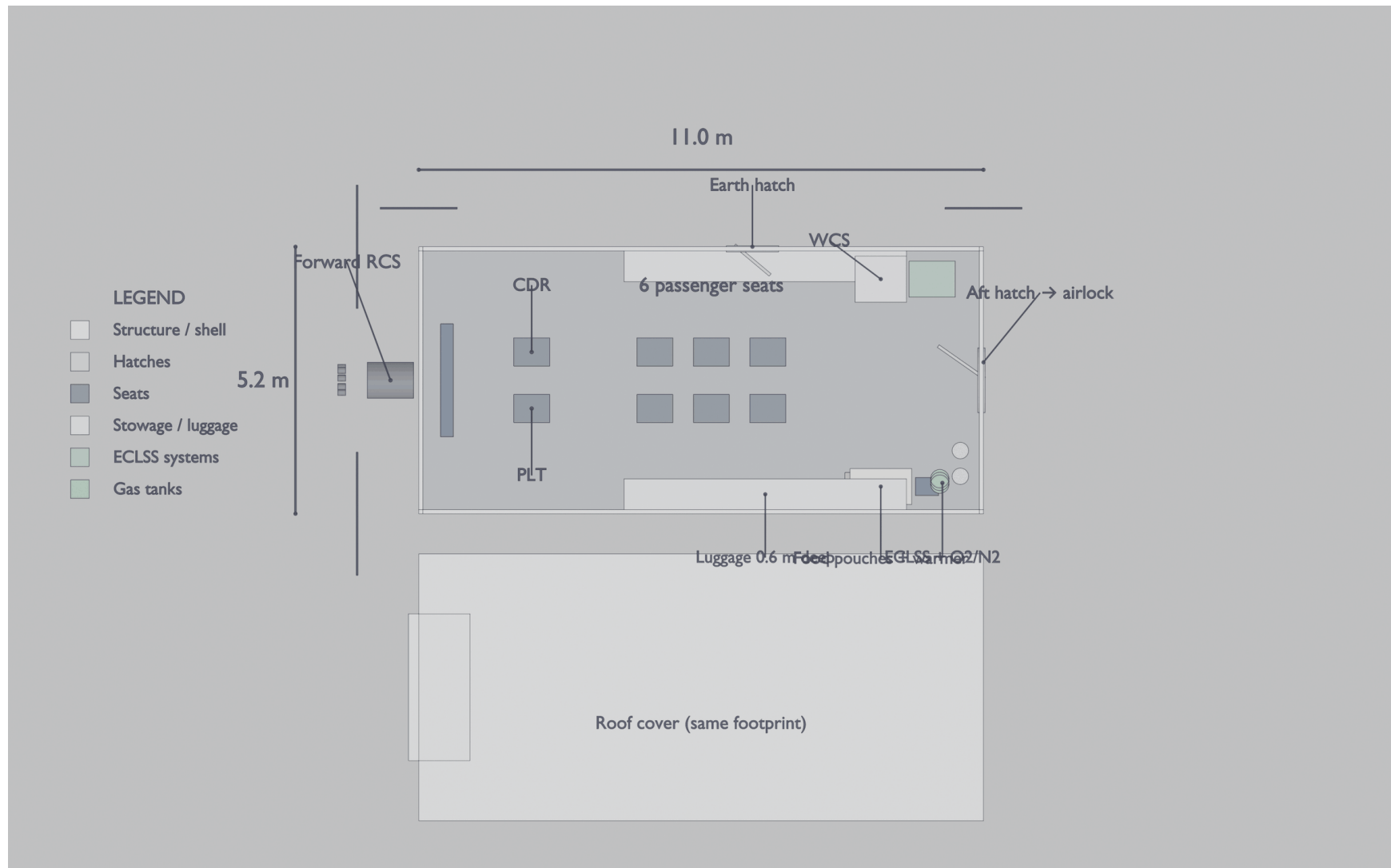


Figure 5 Crew capsule, Blender top-down cutaway from assembly.json.

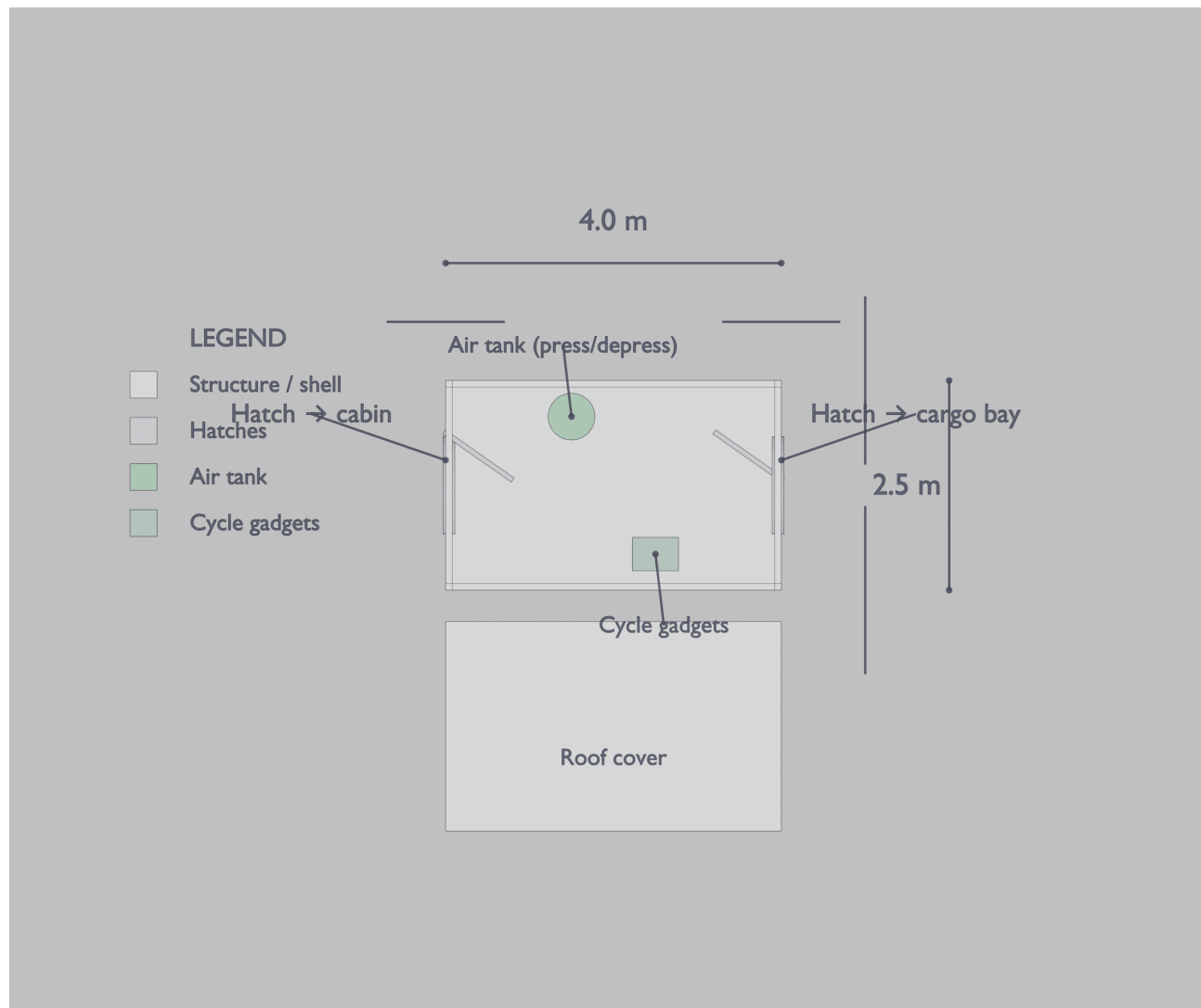


Figure 6 Airlock, Blender top-down cutaway from assembly.json.

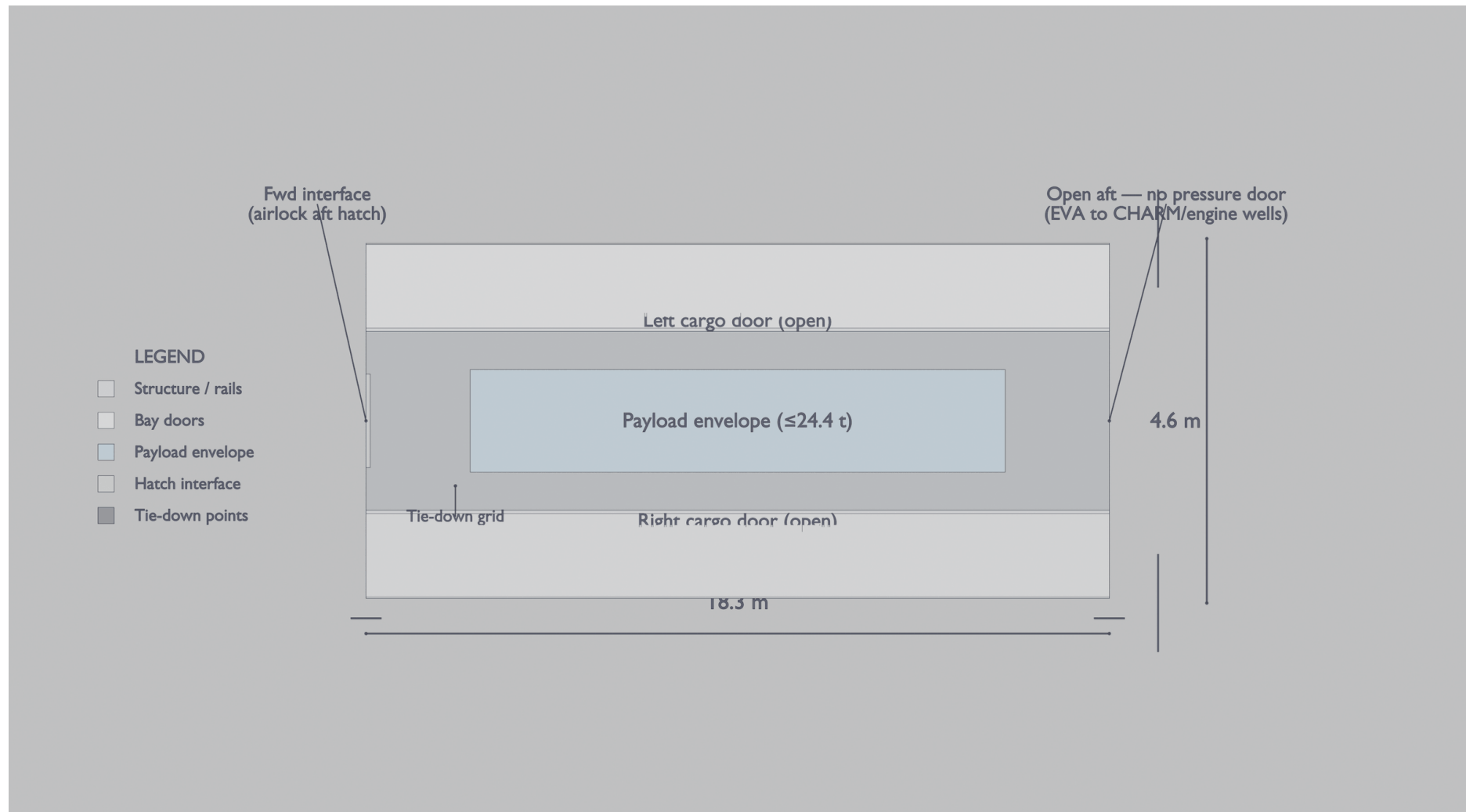


Figure 7 Cargo skid, Blender top-down cutaway from assembly.json with bay doors open.

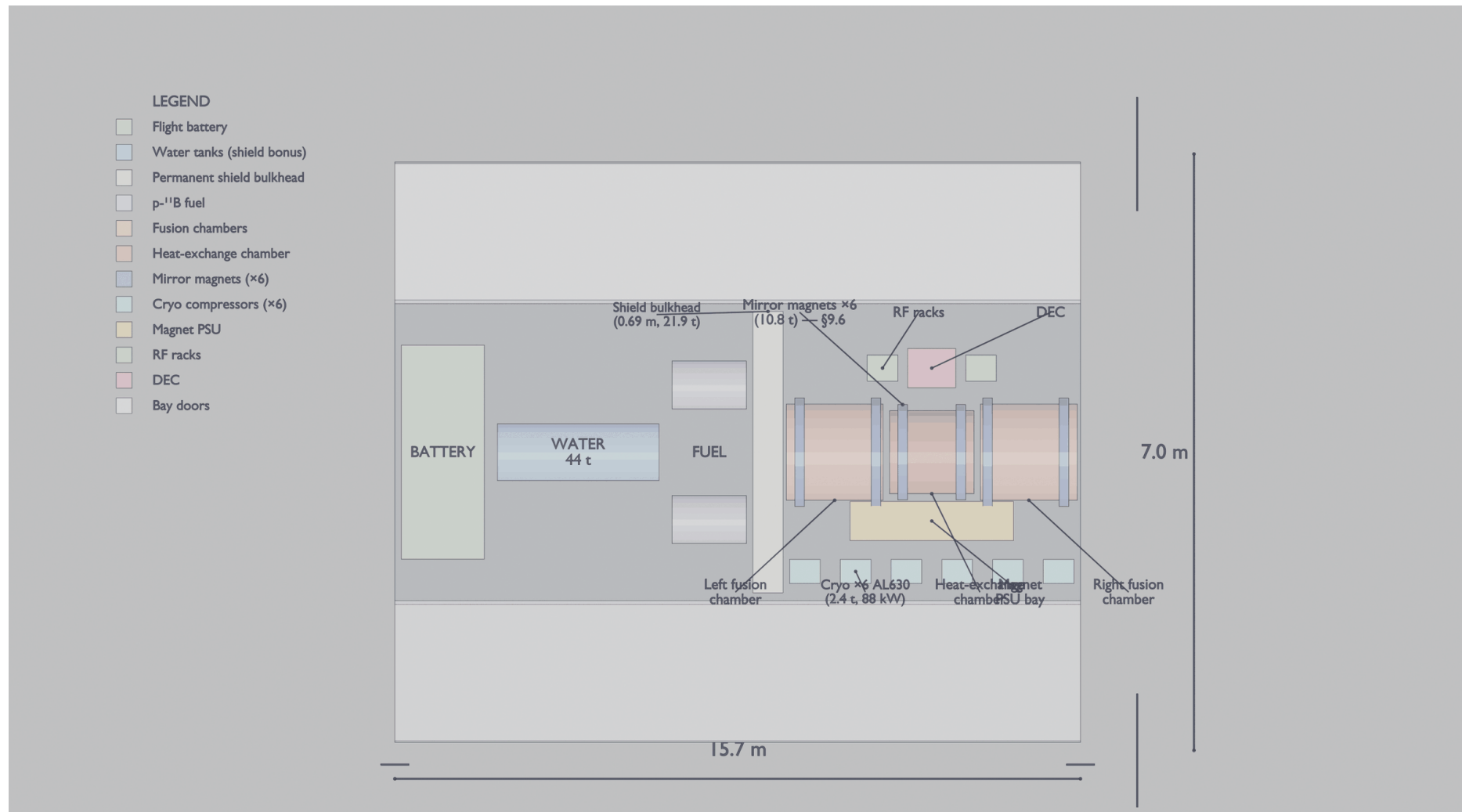


Figure 8 Fusion plant skid, Blender top-down cutaway from assembly.json: flight battery, relocated water tanks (radiation-shield buffer, \$9.3), p-11B fuel, permanent shield bulkhead, chamber string, 6 mirror magnets, 6-unit cryo compressor bay, magnet PSU, RF, and DEC.

1.4 Fusion electric plant (assembly SSOT)

Fig. ~9 is auto-generated from `research/figures/cad/assembly.json` on every build (`scripts/update_arxiv_mermaid.py`, same visible-set algorithm as the interactive outliner — see `research/figures/cad/lib/mermaid_builder.py`). Boxes are plant parts/collections, not a separate physics cartoon; this figure is hard-scoped to the fusion plant assembly, so the one connection leaving it (to the combined-cycle engine) is drawn as a dashed boundary stub rather than pulling in the engine's own parts.

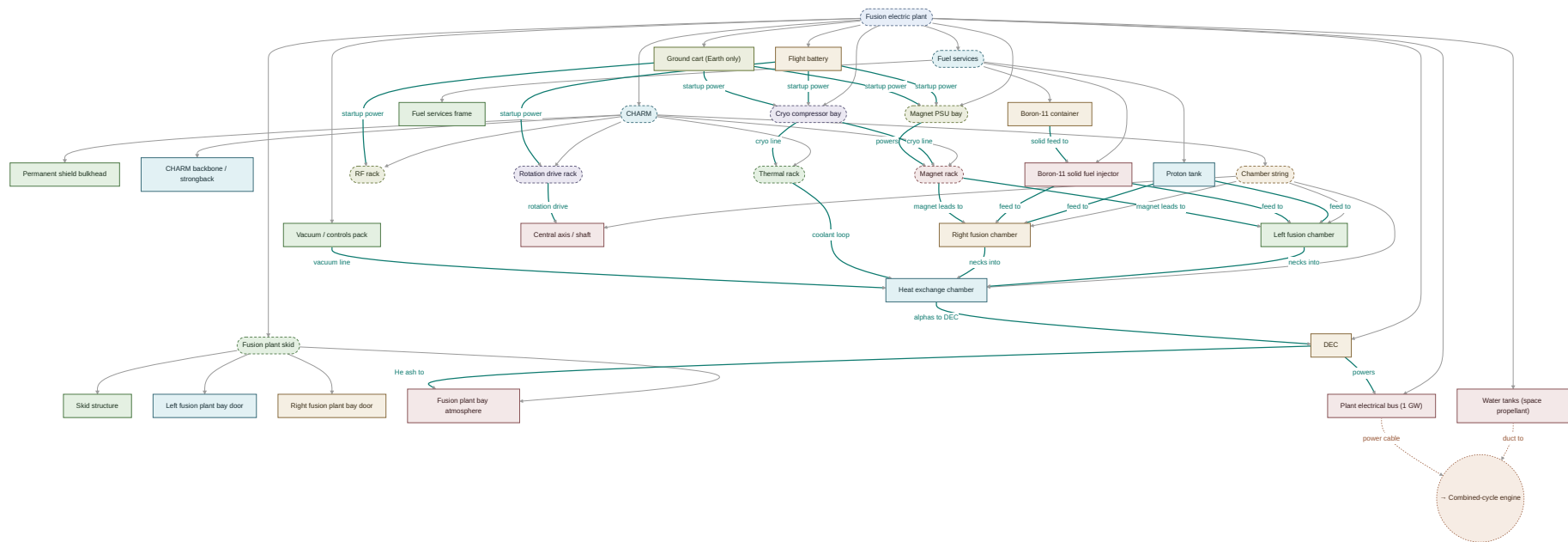


Figure 9 Fusion electric plant (assembly.json).

1.5 Profile stations

Stations match assembly envelopes: crew 0–11 m, airlock 11–15 m, cargo 15–33.3 m, fusion plant 33.3–49 m (battery + water + fuel + CHARM on one skid — water relocated ahead of CHARM as a supplemental shield, §9.3), engine 49–52 m. Fig.~10 is auto-generated (whole-vehicle scope, one level into the fusion plant and engine) by the same `scripts/update_arxiv_mermaid.py` pipeline as Fig.~9.

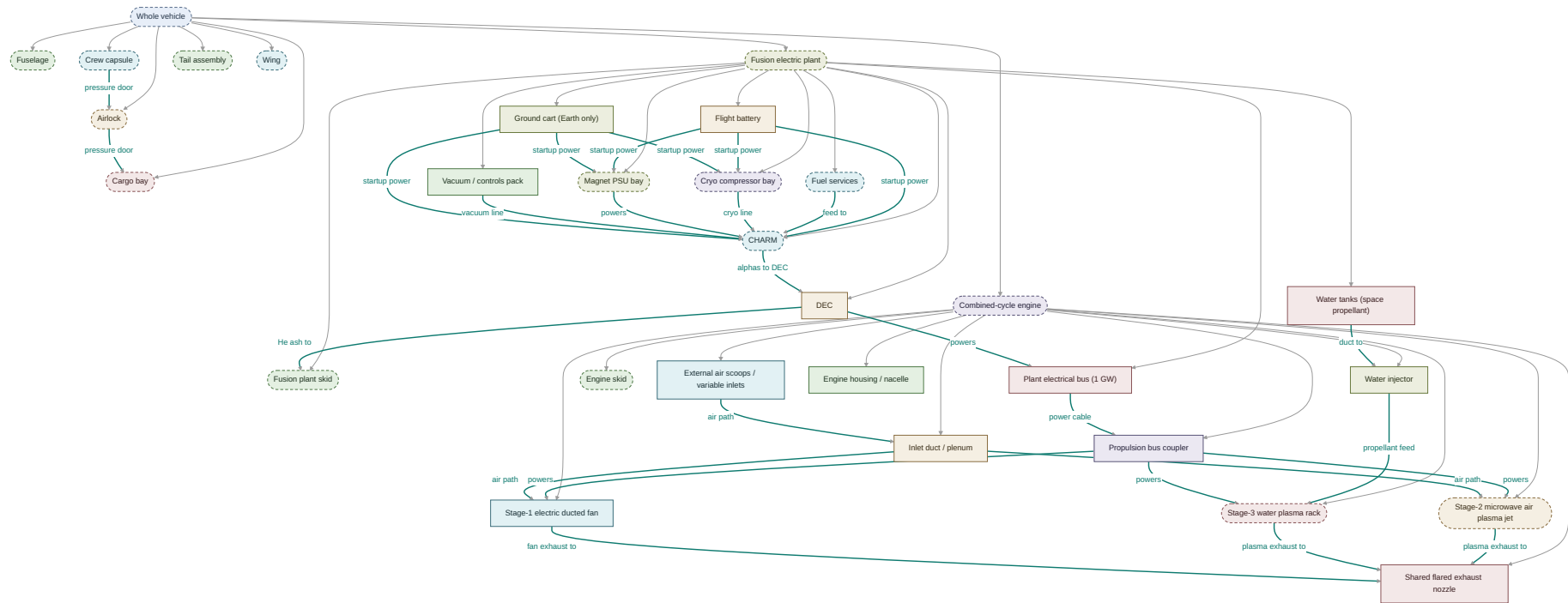


Figure 10 Profile stations from assembly envelopes.

2 Design goals (the plane)

Table 1 Design goals for the plane.

ID	Goal	Statement
G1	SSTO	Single stage from runway to ISS-class LEO; no discarded boosters or external tank
G2	Shuttle style	Orbiter-like airframe: wing-body, thermal protection system (TPS), runway landing, gear, control surfaces; crew systems (toilet, ECLSS, food)
G2b	Single-deck crew	Flatten Shuttle cabin to one deck: flight deck; six reclining seats; O_2/N_2 + ECLSS; luggage; forward/side ground door; large aft airlock into bay
G2c	Length	Stretch OML as needed for airlock, ECLSS tanks, battery, and fusion-fuel tanks—reference $L \approx 52$ m
G3	Shuttle cargo bay	Usable bay ≈ 18.3 m $\times$ 4.6 m class for payload—not filled with reactors
G4	Payload	Shuttle-class cargo: $m_{pl} = 24\,400$ kg reference
G5	Destination	Circular LEO compatible with ISS altitude (≈ 400 km); plane-change to 51.6° treated as margin
G6	Municipal airport	Takeoff/landing on long civil runways; first gear cool enough for noise/oversight; no vertical pad
G7	One engine	Single combined-cycle propulsion string; deadstick glide if plant fails
G8	Clean fuel	$p + {}^{11}\text{B} \rightarrow 3\alpha + 8.7$ MeV; CHARM bottle; DEC-first electricity
G9	Power	Plant electrical bus peak $P_\star = 1$ GW (design target)

Not goals: expendable stages; D–T breeding plant; filling the bay with the fusion island.

3 Constants and symbols

Table 2 Physical constants and reference symbols.

Symbol	Meaning	Reference value
R_E	Earth radius	6.371×10^6 m
μ_E	GM_E	3.986×10^{14} m ³ /s ²
h_{ISS}	ISS-class altitude	4.00×10^5 m

Symbol	Meaning	Reference value
r	Orbit radius	$R_E + h_{\text{ISS}}$
g_0	Standard gravity	9.80665 m/s^2
m_{pl}	Cargo-bay payload	$2.44 \times 10^4 \text{ kg}$
$P_\star$	CHARM bus peak power	$1.00 \times 10^9 \text{ W}$

Masses (all kg):

Table 3 Mass symbol definitions.

Symbol	Meaning
m_{af}	Airframe primary: fuselage, wings, TPS (excl. gear/controls)
m_{gear}	Landing gear (nose + mains), doors, actuators
m_{ctrl}	Control surfaces + actuators (elevons, rudder, speed brake / body flap equiv.)
m_{crew}	Crew cabin systems: ECLSS, O ₂ /N ₂ tanks, pressure control, toilet, galley, food, luggage, airlock fittings
m_{str}	$m_{\text{af}} + m_{\text{gear}} + m_{\text{ctrl}} + m_{\text{crew}}$
m_{eng}	Combined-cycle engine + inlets + nozzles + ducts
m_{C}	CHARM island (chambers, magnets, radio-frequency (RF), DEC, local shield, cryo, bath)
m_{bat}	Flight battery / auxiliary power unit (APU) (restart + hotel)
m_{f}	$p\text{-}^{11}\text{B}$ fuel
m_{w}	Water carried at takeoff (vacuum reaction mass)
m_{dry}	All mass except water
m_0	Gross liftoff mass (GLOW)
m_{ins}	Mass at LEO insertion (after water burn)

4 MWh budget: Shuttle-class mass to ISS LEO

4.1 Ideal orbital specific energy

$$r = R_E + h_{\text{ISS}}, \quad v_{\text{orb}} = \sqrt{\frac{\mu_E}{r}}, \quad \varepsilon_{\text{orb}} = -\frac{\mu_E}{2r}, \quad \varepsilon_{\text{surf}} \approx -\frac{\mu_E}{R_E}.$$

$$\Delta\varepsilon = \varepsilon_{\text{orb}} - \varepsilon_{\text{surf}} = \frac{\mu_E}{R_E} - \frac{\mu_E}{2r} \approx 3.31 \times 10^7 \text{ J/kg} = 33.1 \text{ MJ/kg}.$$

Numerical anchors:

$$v_{\text{orb}} \approx 7.67 \text{ km/s}, \quad \Delta\varepsilon \approx 9.19 \text{ kWh/kg}.$$

4.2 Orbital energy of the inserted vehicle

Everything that arrives at ISS altitude is still aboard (SSTO):

$$E_{\text{orb}} = m_{\text{ins}} \Delta\varepsilon.$$

In MWh:

$$E_{\text{orb}}^{(\text{MWh})} = m_{\text{ins}} \cdot \frac{\Delta\varepsilon}{3.6 \times 10^9}.$$

Table 4 Orbital energy for representative inserted masses.

m_{ins}	E_{orb}	E_{orb}
100 t (orbiter+cargo class)	3.31 TJ	920 MWh
150 t	4.97 TJ	1380 MWh
200 t	6.62 TJ	1840 MWh

4.3 Source energy (plant must supply)

Air-breathing and rocket/plasma paths leave energy in the wake and fight drag/gravity. Define a mission multiplier $\kappa_E \geq 1$:

$$E_{\text{src}} = \kappa_E E_{\text{orb}} = \kappa_E m_{\text{ins}} \Delta\varepsilon.$$

Working band for planning:

$$\kappa_E \in [2, 4] \Rightarrow E_{\text{src}}^{(\text{MWh})} \approx (18\text{--}37) m_{\text{ins},100\text{t}}$$

i.e. about 1.8–3.7 GWh of source energy per 100 t inserted, scaling linearly with m_{ins} .

Time at constant bus power $P_\star$:

$$t_{\text{src}} = \frac{E_{\text{src}}}{P_\star} = \frac{\kappa_E m_{\text{ins}} \Delta\varepsilon}{P_\star}.$$

For $m_{\text{ins}} = 1.9 \times 10^5$ kg, $\kappa_E = 3$, $P_\star = 1$ GW:

$$E_{\text{src}} \approx 18.9 \text{ TJ} \approx 5240 \text{ MWh}, \quad t_{\text{src}} \approx 5.2 \text{ h}.$$

So a 1 GW CHARM bus is an **energy-throughput** match for Shuttle-class SSTO only if the climb/insert lasts hours at high average power—or peak $P_\star$ is used with κ_E toward the low end and a lighter m_{ins} .

ISS note. Matching ISS inclination adds plane-change Δv . Fold into vacuum Δv margin (§5) rather than into $\Delta\varepsilon$.

5 Flight regimes: three electric stages

One propulsion string, three stages. No scramjet claim. Plant couples only by **power cable** (DEC $\rightarrow$ bus).

Table 5 Combined-cycle stages (reaction mass and thruster family).

Stage	Ambient	Reaction mass	Thruster family (anchor lit.)
1 Munici- pal / dense air	Free air	Ingested air	Electric ducted fan (EDF)
2 Climb / scarce air	Free air	Ingested + compressed air	Microwave air plasma jet [14]
3 Vacuum / LEO insert	Intakes sealed	Carried water	Microwave water plasma thruster lineage [15]

Cutoff: intakes seal at density/Mach command $\rho \leq \rho_{\text{seal}}$ (or crew/auto seal for reentry); stage 3 begins.

Energy split (schematic):

$$E_{\text{src}} = E_1 + E_2 + E_3 + E_{\text{hotel}} + E_{\text{loss}}.$$

With free reaction mass in stages 1–2, plant energy primarily raises vehicle mechanical energy and pays drag:

$$P_{\text{prop}}(t) \approx \frac{D(t) v(t)}{\eta_p(t)}, \quad \int P_{\text{prop}} dt \subset E_1 + E_2.$$

In stage 3, jet power for exhaust speed v_e and water mass flow $\dot{m}_w$ is

$$P_{\text{jet}} = \frac{1}{2} \dot{m}_w v_e^2, \quad T = \dot{m}_w v_e \quad \Rightarrow \quad P_{\text{jet}} = \frac{1}{2} T v_e.$$

6 Water mass as a function of whole dry mass

Water is used only in stage 3. Let

$$m_{\text{dry}} = m_{\text{str}} + m_{\text{pl}} + m_{\text{eng}} + m_{\text{C}} + m_{\text{bat}} + m_{\text{f}},$$

$$m_0 = m_{\text{dry}} + m_{\text{w}}, \quad m_{\text{ins}} = m_{\text{dry}}$$

(all water expended by insertion). Rocket equation for vacuum Δv_{vac} :

$$\Delta v_{\text{vac}} = v_e \ln \left(\frac{m_{\text{dry}} + m_{\text{w}}}{m_{\text{dry}}} \right) = v_e \ln \mu, \quad \mu = e^{\Delta v_{\text{vac}}/v_e}.$$

Water constraint (closed form):

$$m_{\text{w}} = m_{\text{dry}} \left(e^{\Delta v_{\text{vac}}/v_e} - 1 \right) = m_{\text{dry}} (\mu - 1)$$

$$m_0 = m_{\text{dry}} \mu = m_{\text{dry}} e^{\Delta v_{\text{vac}}/v_e}$$

Vacuum Δv budget:

$$\Delta v_{\text{vac}} = v_{\text{orb}} - v_{\text{ab}} + \Delta v_{g,\text{vac}} + \Delta v_{\text{steer}} + \Delta v_{\text{ISS plane}}.$$

Reference planning values:

$$v_{\text{ab}} = 3.5 \text{ km/s}, \quad \Delta v_{g,\text{vac}} + \Delta v_{\text{steer}} = 0.8 \text{ km/s}, \quad \Delta v_{\text{ISS plane}} = 0.2 \text{ km/s} \Rightarrow \Delta v_{\text{vac}} \approx 5.2 \text{ km/s}.$$

Optimistic air-breathing ($v_{\text{ab}} = 5 \text{ km/s}$, small margins) $\rightarrow \Delta v_{\text{vac}} \approx 3.5\text{--}4 \text{ km/s}$.

Exhaust speed from specific impulse:

$$v_e = I_{\text{sp}} g_0.$$

Table 6 Water mass fraction versus vacuum Δv and I_{sp} .

Δv_{vac}	I_{sp}	v_e	μ	$m_{\text{w}}/m_{\text{dry}}$
4 km/s	2000 s	19.6 km/s	1.226	0.226
4 km/s	3000 s	29.4 km/s	1.146	0.146
5.2 km/s	2000 s	19.6 km/s	1.302	0.302
5.2 km/s	5000 s	49.0 km/s	1.112	0.112

7 Vehicle sizing equations

7.1 Structure, crew, gear, controls, and bay

Cargo bay geometry (goal G3):

$$L_{\text{bay}} = 18.3 \text{ m}, \quad D_{\text{bay}} = 4.6 \text{ m}, \quad V_{\text{bay}} \approx \frac{\pi}{4} D_{\text{bay}}^2 L_{\text{bay}} \approx 304 \text{ m}^3.$$

Payload density check:

$$\bar{\rho}_{\text{pl}} = \frac{m_{\text{pl}}}{V_{\text{bay}}} \approx 80 \text{ kg/m}^3$$

(compatible with mixed cargo; bay remains payload volume).

Crew (Shuttle functions, single deck). Forward **flight deck**: commander and pilot **facing forward** into windows and a full control-panel wall [12,13]. Living volume: **six forward-facing passenger seats** (Crew Dragon-like rows, stretched cabin) plus the flight-deck pair; **waste collection system (WCS)**; **galley/food station without a kitchen sink** (0g); **crew luggage** with doors into the aisle; **ECLSS** with **O₂/N₂ tankage inside the pressure vessel**. **Solid port side hatch** (Earth/runway only) and **solid aft pressure hatch** to the airlock. **Airlock** oversized vs a suit-closet: dual-hatch volume on the aft cabin bulkhead facing the **cargo bay**, sized for suited egress (Shuttle middeck airlock pattern, not undersized) [13].

Landing gear and control surfaces are explicit mass lines (not buried only in a lump “structure” number):

$$\begin{aligned} m_{\text{af}} &= 7.20 \times 10^4 \text{ kg} && \text{(longer fuselage, wings, TPS, doors),} \\ m_{\text{gear}} &= 4.00 \times 10^3 \text{ kg} && \text{(nose + dual main trucks; mass bill only—see profile),} \\ m_{\text{ctrl}} &= 3.00 \times 10^3 \text{ kg} && \text{(elevons, rudder, speed-brake equiv. + actuators),} \\ m_{\text{crew}} &= 8.50 \times 10^3 \text{ kg} && \text{(ECLSS, O}_2\text{/N}_2\text{, WCS, galley, food, luggage, airlock),} \\ m_{\text{str}} &= m_{\text{af}} + m_{\text{gear}} + m_{\text{ctrl}} + m_{\text{crew}} = 8.75 \times 10^4 \text{ kg.} \end{aligned}$$

CHARM and propulsion water **do not** consume V_{bay} . Plant volume sits in a **fuselage island** (spine / aft of bay / wing carry-through), subject to

$$V_C \leq V_{\text{island}}^{\text{max}} \approx 80\text{--}150 \text{ m}^3 \quad (\text{guesstimate for pancaked Shuttle envelope}).$$

7.2 CHARM mass hole

Define island specific power on the **electrical bus**:

$$\alpha_C = \frac{P_\star}{m_C} \quad \Rightarrow \quad \boxed{m_C = \frac{P_\star}{\alpha_C}}.$$

Table 7 CHARM island mass versus specific power at 1 GW.

α_C	m_C at $P_\star = 1 \text{ GW}$	Comment
5 kW/kg	200 t	Heavy; fights SSTO
10 kW/kg	100 t	Stretch
15 kW/kg	67 t	Reference hole — bottom- up roll-up in §9.3
25 kW/kg	40 t	Aggressive

Volume consistency:

$$\bar{p}_C = \frac{P_\star}{V_C} \quad \Rightarrow \quad V_C = \frac{P_\star}{\bar{p}_C}.$$

For $V_C \leq 120 \text{ m}^3$: $\bar{p}_C \geq 8.3 \text{ MW/m}^3$ bus-averaged over the island—severe packaging.

7.3 Engine and battery holes

$$m_{\text{eng}} = m_{\text{EDF}} + m_{\mu\text{air}} + m_{\text{wth}} + m_{\text{shared}} = 1.5 \times 10^4 \text{ kg} \quad (\text{reference packaging hole; closed in §10.2}),$$

$$m_{\text{bat}} = 2.0 \times 10^3 \text{ kg} \quad (\text{restart} + \text{hotel; ground cart does first light}),$$

$$m_{\text{f}} = 5.0 \times 10^2 \text{ kg} \quad (p\text{-}^{11}\text{B} + \text{H inventory; water dominates expendables; §10.2}).$$

7.4 Closed dry / wet mass

$$\begin{aligned}
m_{\text{str}} &= m_{\text{af}} + m_{\text{gear}} + m_{\text{ctrl}} + m_{\text{crew}}, \\
m_{\text{dry}} &= m_{\text{str}} + m_{\text{pl}} + m_{\text{eng}} + \frac{P_{\star}}{\alpha_{\text{C}}} + m_{\text{bat}} + m_{\text{f}}, \\
m_{\text{w}} &= m_{\text{dry}} \left(e^{\Delta v_{\text{vac}}/v_e} - 1 \right), \\
m_0 &= m_{\text{dry}} e^{\Delta v_{\text{vac}}/v_e}, \\
m_{\text{ins}} &= m_{\text{dry}}.
\end{aligned}$$

8 Solved reference vehicle (all-up mass)

Freeze:

$$\begin{aligned}
P_{\star} &= 1 \text{ GW}, \quad \alpha_{\text{C}} = 15 \text{ kW/kg}, \quad \Delta v_{\text{vac}} = 4.0 \text{ km/s}, \\
I_{\text{sp}} &= 2000 \text{ s}, \quad v_e = I_{\text{sp}} g_0 = 19.61 \text{ km/s}.
\end{aligned}$$

Solve:

$$m_{\text{C}} = \frac{10^9}{1.5 \times 10^4} = 6.67 \times 10^4 \text{ kg} \quad (\text{bottom-up roll-up: §9.3}).$$

$$m_{\text{str}} = 72\,000 + 4\,000 + 3\,000 + 8\,500 = 8.75 \times 10^4 \text{ kg},$$

$$m_{\text{dry}} = 87\,500 + 24\,400 + 15\,000 + 66\,667 + 2\,000 + 500 = 1.961 \times 10^5 \text{ kg} \quad (196.1 \text{ t}),$$

$$\mu = e^{4000/19613} = 1.226, \quad m_{\text{w}} = 0.226 m_{\text{dry}} = 4.44 \times 10^4 \text{ kg} \quad (44.4 \text{ t}),$$

$$m_0 = 2.404 \times 10^5 \text{ kg} \quad (\mathbf{240 \text{ t GLOW}}),$$

$$m_{\text{ins}} = 196.1 \text{ t}.$$

Mass bill (reference):

Table 8 Reference vehicle mass bill.

Item	Mass
Airframe + TPS (longer OML)	72.0 t
Landing gear	4.0 t
Control surfaces + actuators	3.0 t
Crew systems (ECLSS, O ₂ /N ₂ , WCS, galley, food, luggage, airlock)	8.5 t
Payload (cargo bay)	24.4 t
Combined-cycle engine (EDF 5.0 + air-plasma 4.4 + water thruster 3.1 + shared 2.5)	15.0 t
CHARM island (bottom-up roll-up: magnets + cryo + unsized remainder, §9.3)	66.7 t
Flight battery	2.0 t
$p\text{-}^{11}\text{B}$ / proton fuel (+ tankage)	0.5 t

Item	Mass
Water (vacuum propellant)	44.4 t
GLOW	240 t

Energy at this mass:

$$E_{\text{orb}} = 6.49 \text{ TJ} = 1800 \text{ MWh}, \quad E_{\text{src}}(\kappa_E = 3) = 5410 \text{ MWh}, \quad t_{\text{src}}(1 \text{ GW}) = 5.4 \text{ h}.$$

Sensitivity (same Δv_{vac} , I_{sp} ; m_{str} fixed):

Table 9 Sensitivity of dry and wet mass to specific power.

α_C	m_C	m_{dry}	m_{w}	m_0
10 kW/kg	100 t	229 t	52 t	281 t
15 kW/kg	67 t	196 t	44 t	240 t
25 kW/kg	40 t	169 t	38 t	208 t

Municipal takeoff weight $\sim 210\text{--}280$ t is heavy vs airliners but in the large-military / 747-class band—not a Citation.

9 Constraints on the CHARM power plant (summary)

This section states only the vehicle-level requirement vector $\mathcal{R}_C$ that the CHARM plant must satisfy and the resulting mass/power numbers this paper’s mass closure (§7–§8) depends on. The full bottom-up derivation — magnet/cryo mass roll-up anchored to real WHAM/SPARC hardware, the permanent radiation-shield bulkhead sizing, light-off sequence, and the itemized unobtainium list — now lives in the companion reactor paper, “**Compact CHARM p-¹¹B fusion reactor for electricity generation for aerospace propulsion**” [16]; see that paper for citations, sensitivity cases, and the honestly-flagged unsized remainder.

9.1 Power, mass, and volume

$$P_{\text{bus}}(t) \geq P_{\text{prop}}(t) + P_{\text{hotel}}(t), \quad \max P_{\text{bus}} = P_{\star} = 1 \text{ GW},$$

$$m_C = \frac{P_{\star}}{\alpha_C}, \quad \alpha_C \geq 15 \text{ kW/kg (reference; } \geq 10 \text{ kW/kg hard floor for SSTO)}.$$

$$V_C \leq 120 \text{ m}^3, \quad \bar{p}_C = P_{\star}/V_C \geq 8 \text{ MW/m}^3.$$

9.2 Fuel, ash, DEC, restart, and duty (one line each)

- **Fuel/ash:** continuous $p\text{-}^{11}\text{B}$ burn, $m_{\text{f}}(t_{\text{mission}}) \ll m_{\text{w}}$; ash (He) strained per CHARM’s multi-chamber design.
- **DEC:** $\eta_{\text{DEC}} \gtrsim 0.4\text{--}0.7$ on ordered α /wave channel; thermal reject dumped to the air path, an island bath/turbine, or (exo-atmospheric only) radiators.

- **Space restart:** $E_{\text{restart}} \leq 300\text{--}500\text{ kWh}$ from a $m_{\text{bat}} = 2\text{ t}$ battery — enough for pilot-chamber relight, not for ascent; first light on Earth uses a ground cart, not carried.
- **Continuous operation:** recirculating power (RF walls, rotation, magnets, vacuum, cryo) is designed small enough that the 1 GW bus does not require a multi-MW neutral-beam-injection (NBI) farm [1,8,9].
- **Municipal/flight safety:** no tritium breeding inventory; ramp/cabin doses must meet civil constraints in fan mode; single-string plant — engine-out $\equiv$ plant-out $\rightarrow$ glide.

9.3 CHARM island mass: magnets, cryo, and shielding (reference numbers)

The reactor paper [16] sizes the CHARM island bottom-up against real hardware anchors — the Wisconsin HTS Axisymmetric Mirror (WHAM) magnets and the SPARC Toroidal Field Model Coil cryo program — plus a permanent radiation-shield bulkhead, and carries an explicit, honestly-labeled unsized remainder for RF hardware and chamber/backbone structure. The reference numbers this paper’s mass closure uses are:

Table 10 CHARM island reference mass/power (full derivation: [16]).

Item	Mass / power
Magnets (6, WHAM-anchored)	$\approx 10.8\text{ t}$
Cryo compressor bay (6 flight-remanufactured AL630s)	$\approx 2.4\text{ t}, \approx 88\text{ kW}$
Permanent radiation-shield bulkhead (polyethylene, sized for empty water tank)	$\approx 21.9\text{ t} (0.69\text{ m})$
RF hardware + backbone/chamber structure	$\approx 31.5\text{ t}$, not independently sized — tracked unobtainium
CHARM island total	66.7 t $\Rightarrow \alpha_{\text{C,implied}} = 15.00\text{ kW/kg}$
$m_{\text{C}} =$	
$\max(m_{\text{C,target}}, m_{\text{bottom-up}})$	

Because the known bottom-up pieces fit inside the 67 t top-down target, m_{C} , m_{dry} , and GLOW in §8 do not move — but a future RF/shield/structure sizing pass (or a heavier magnet/cryo number) in the reactor paper would cascade into m_0 automatically. A full-tank water slab (§11, relocated ahead of CHARM) supplements the permanent bulkhead with $\gg 30\text{ dB}$ of additional photon/neutron attenuation when present; RF/microwave leakage is a separate, near-zero-incremental-mass Faraday-cage problem. See [16] §9 for the shared shielding methodology, sensitivity cases, and the flight-cryocooler ceiling check.

10 Combined-cycle engine (summary)

This section states only the vehicle-level stage architecture and the closed powers/masses that this paper’s mass closure (§7–§8) depends on. The full derivation — VSPAERO/OpenFOAM/SU2

aero checks, the Stage-2 constant- Q climb integrator, Stage-3 water/ I_{sp} trade cases, the acoustic-signature calibration, and the orbital ascent profile — now lives in the companion engine paper, “**Three cycle electric SSTO rocket engine**” [17]; see that paper for citations, pass/fail smoke-test tables, and the itemized propulsion unobtainiums.

One propulsion string with stage index $\sigma \in \{1, 2, 3\}$. Plant couples **only** by power cable. Stages 1–2 burn **free air** (reaction mass not carried); stage 3 burns **carried water**. Fusion fuel m_f is not propellant for the nozzle.

10.1 Stage map (reference)

Table 11 Stages, reaction mass, and closed reference performance at the solved vehicle (§8; full derivation [17]).

σ	Name	Reaction mass	Peak power	Duration	Energy
1	Electric ducted fan	Free air	$P_1^* \approx 93 \text{ MW}$	133 s	3.4 MWh
2	Microwave air plasma jet	Free compressed air	$P_2^* = 995 \text{ MW}$	28.7 min	476 MWh
3	Water plasma thruster	Carried H_2O (44 t, $I_{\text{sp}} = 2000 \text{ s}$)	$P_3^* = 995 \text{ MW}$	4.33 h	4309 MWh
Hotel	Plant recirculating floor	—	5 MW	$t_1 + t_2 + t_3$	24.2 MWh

$P_2^* = P_3^*$ is a bus-ceiling coincidence, not physics: stage 2 is a short, low-mass-flow hypersonic climb; stage 3 is a long, power-limited vacuum burn that expends the entire water store — [17] derives t_2 , h_{seal} , and $E_2 \ll E_3$ independently from a constant-ascent-dynamic-pressure climb integrator, then reconciles the bottom-up stage-energy total ($\approx 4812 \text{ MWh}$) against the top-down §4/§8 $\kappa_E \in [2, 4]$ budget ($\kappa_{E,\text{implied}} \approx 2.67$).

Switching between stages is by atmospheric density and Mach number:

$$\sigma = \begin{cases} 1 & \rho > \rho_{12}, \quad M < M_{12}, \\ 2 & \rho > \rho_{\text{seal}}, \quad M \geq M_{12}, \\ 3 & \rho \leq \rho_{\text{seal}} \text{ or intakes sealed.} \end{cases}$$

10.2 Engine mass budget

Table 12 Engine component mass allocation and implied packaging specific power (reference hole $m_{\text{eng}} = 15 \text{ t}$; full sizing/literature anchors [17]).

Component	Mass	Sized to	Implied $\alpha = P/m$
Stage-1 EDF (motor+fan+duct)	5.0 t	P_1^*	$\sim 13 \text{ kW/kg}$ (near NASA HEMM-class motors)

Component	Mass	Sized to	Implied $\alpha = P/m$
Stage-2 MW farm + applicator + precompress	4.4 t	P_2^*	$\sim 230 \text{ kW/kg}$ (packaging unobtainium)
Stage-3 thruster head + vaporizer/feed	3.1 t	P_3^*	$\sim 320 \text{ kW/kg}$ (packaging unobtainium)
Shared nacelle / nozzle / inlets / bus coupler	2.5 t	structure	—
Engine total	15 t	§8 freeze	—

If stage-2/3 hardware were packaged at a more literal $\sim 8 \text{ kW/kg}$, m_{eng} would jump to $\mathcal{O}(100 \text{ t})$ and GLOW to $\sim 400 \text{ t}$. The 15 t engine line is therefore a **same-class hole as** $\alpha_C = 15 \text{ kW/kg}$ (§13) — while **power** and **water** closes are on firmer ground (rocket-equation water mass and the E_3 invariant both close in closed form; [17] §10.5).

Fusion fuel (not nozzle propellant): stoichiometric $p\text{-}^{11}\text{B}$ rest-mass for the mission bus energy is $\ll 1 \text{ kg}$ at ideal conversion, $\sim 1 \text{ kg}$ at realistic chain efficiency; freeze $m_f = 0.5 \text{ t}$ covers tankage, residuals, and margin — **water dominates expendables**.

10.3 Physical envelope

One nacelle with external air scoops feeding an inlet duct/plenum; stage-1 EDF sits in-duct behind the scoops; a shared flared aft nozzle is the common exit for all three stages. Water tanks sit on the fusion plant skid ahead of CHARM (§9.3, §11) rather than aft of the engine, doubling as a supplemental radiation shield when full. Atmospheric acoustic signature, sonic-boom crossing altitude, and the full ascent-to-orbit trajectory are engine-paper content [17].

11 Layout details

Station map for the vision figures in §1. The **top-down bay connectivity** diagram is a reading aid for the floorplan; gear remains on the exterior profile only.

Table 13 Longitudinal station and bay layout.

Station (m)	Bay	Contents
0–11	Crew module	Forward-facing CDR/PLT flight deck; six forward-facing passenger seats; WCS; galley (no sink); luggage stowage ; ECLSS + O₂/N₂ inside pressure vessel; solid side + aft hatches
—	Ground door	Forward/port crew door (side hatch) — terrestrial ingress only

Station (m)	Bay	Contents
11–15	Airlock	Suited-crew airlock (~ 2.5 m class clear), aft bulkhead facing into cargo bay
15–33.3	Cargo	18.3 m $\times$ 4.6 m payload bay (no reactors)
33.3–35.5	Battery	Flight battery ≈ 2 t (restart / hotel)
35.5–39.5	Water	≈ 44 t H ₂ O — relocated ahead of CHARM: supplemental radiation shield when full (§9.3)
39.5–41.5	Fusion fuel	Proton / ¹¹ B feed tanks + plumbing (low mass, real volume)
41.5–49	CHARM	Reactor island incl. permanent shield bulkhead ($\lesssim 120$ m ³ , 66.7 t)
49–52	Engine	Combined-cycle nacelle + nozzle
Wings	Controls	Elevons, rudder (gear not drawn on this figure)

Doors (Shuttle pattern). (1) **Side/forward crew door** — runway/ground only; (2) **airlock** — on-orbit cabin $\leftrightarrow$ cargo bay / vacuum for suited operations [13]. Fig.~11 is auto-generated (whole-vehicle scope; crew capsule expanded to system level, airlock/cargo bay as single boxes, plant/engine one level) by the same pipeline as Figs.~9–10.

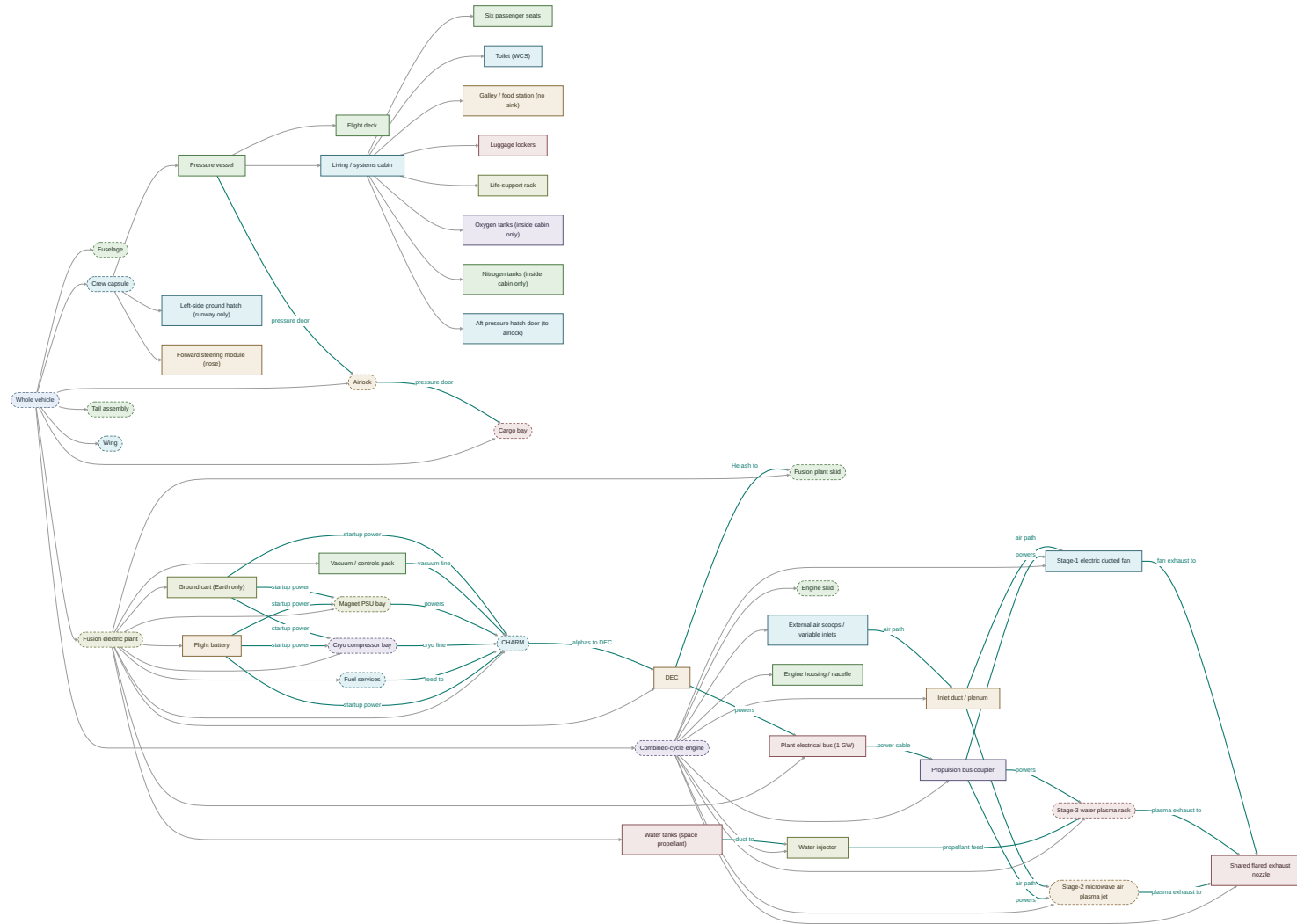


Figure 11 Top-down floorplan from assembly.json.

12 Systems checklist (equations $\rightarrow$ constraints)

Table 14 Systems checklist: equations to constraints.

Step	Governing relations	Binding output
Goals	G1–G9 + single-deck crew	Shuttle bay + municipal SSTO to ISS
LEO energy	$E_{\text{orb}} = m_{\text{ins}}\Delta\varepsilon$, $E_{\text{src}} = \kappa_E E_{\text{orb}}$	~ 1.8 GWh orbital @ 196 t; ~ 5.4 GWh source @ $\kappa_E = 3$
Regimes	Stages 1–3 mass-flow logic	EDF $\rightarrow$ microwave air plasma $\rightarrow$ water plasma
Water	$m_{\text{w}} = m_{\text{dry}}(e^{\Delta v_{\text{vac}}/v_e} - 1)$	$\sim 23\%$ of dry mass @ ref
Structure	$m_{\text{str}} = m_{\text{af}} + m_{\text{gear}} + m_{\text{ctrl}} + m_{\text{crew}}$	87.5 t incl. gear/controls/ECLSS+O ₂ /luggage/airlock
CHARM	$m_C = P_{\star}/\alpha_C$, DEC, restart	67 t @ 15 kW/kg
Light-off	Magnets + RF + rotation; no NBI	50–200 kWh est.; cart / 2 t battery
Solve	$m_0 = m_{\text{dry}}e^{\Delta v_{\text{vac}}/v_e}$	$m_0 \approx 240$ t; $L \approx 52$ m
Engine	$P_1^{\star} \approx 65$ MW; $P_2^{\star} = P_3^{\star} \approx 995$ MW; $m_{\text{w}}(I_{\text{sp}})$	§10.1

13 Imputed CHARM plant specs, gap to present design, and unobtainiums

13.1 Specs this vehicle imputes to CHARM

The SSTO solve does not invent new plasma physics; it **back-solves** a power island that must fit the airframe. Reference imputed plant:

Table 15 Imputed CHARM plant requirements for this SSTO.

Quantity	Imputed requirement	Source in this note
Fuel	Continuous $p\text{-}^{11}\text{B}$	G8; §2
Architecture	Multi-chamber rotating mirror; species separation; ash strain; DEC	§3, §9, §1; [1,8–11]
Electrical bus peak	$P_{\star} = 1$ GW	G9; Phase B/C power
Mission source energy	~ 5 GWh class per ascent ($\kappa_E \sim 3$)	§4, §8
Island mass	$m_C \approx 67$ t	$\alpha_C = 15$ kW/kg
Specific power (bus)	$\alpha_C \geq 15$ kW/kg (floor ~ 10 kW/kg)	§7.2, §9.1
Island volume	$V_C \lesssim 120$ m ³	Fuselage bay aft of cargo
Volumetric power	$\bar{p}_C \gtrsim 8$ MW/m ³	$P_{\star}/V_C$

Quantity	Imputed requirement	Source in this note
DEC	$\eta_{\text{DEC}} \sim 0.4\text{--}0.7$ on ordered α / wave channel	§9.2
Aux heating	RF + rotation + magnets; no multi-MW NBI farm	§9.2
Magnets (6, WHAM-anchored)	$\approx 10.8\text{ t}$ (16.2% of m_C)	§9.3
Cryo compressor bay (6 AL630, flight-remanufactured)	$\approx 2.4\text{ t}$, $\approx 88\text{ kW}$ (3.6% of m_C)	§9.3
Permanent radiation shield bulkhead (poly, sized empty-tank)	$\approx 21.9\text{ t}$ (0.69 m), first sourced piece of the remainder	§9.3
RF + back-bone/chamber structure	$\approx 31.5\text{ t}$ remaining, not independently sized	§9.3
Light-off energy	$\sim 50\text{--}200\text{ kWh}$ to useful plasma (est.)	§9.2
Space restart	Pilot-string from $\sim 2\text{ t}$ battery ($\sim 300\text{--}500\text{ kWh}$)	§6, §9.2
Duty	Continuous burn through climb/insert; throttleable bus	§5
Environment	Flight loads, TPS-adjacent thermal, municipal dose with fan-mode plant	G6, §9.2

13.2 Where published CHARM stands today

Relative to that invoice, the Fisch / Advanced Research Projects Agency–Energy (ARPA-E) / Pale Blue line today is a **physics and intellectual-property (IP) program**, not a flight power plant [1,2]:

Table 16 Gap between present CHARM and vehicle need.

Gate	Present CHARM (public)	Vehicle need
Fuel / kinetics	Strong papers on hybrid fast–thermal $p\text{--}^{11}\text{B}$, alpha channeling, ash poisoning [3–7]	Same fuel family — aligned in intent
Chambering	Architecture + patent filings for separated reactants, ponderomotive walls, open-trap HV, differential confinement [1,8–11]	Must work together in one island
Lawson / Q	Component studies and 0D balances; integrated self-consistent power-positive reactor not demonstrated [1]	Net bus power after recirculating RF/rotation
Hardware	No public GW-class (or even pilot) CHARM machine; software-first / early company path	Flyable 67 t island
DEC	Themes in theory and open-trap high-voltage (HV) patents [1,10]	Flight-qualified η_{DEC} at GW

Gate	Present CHARM (public)	Vehicle need
Mass / volume	Not published as kW/kg or MW/m ³ plant envelopes	15 kW/kg, $\gtrsim$ 8 MW/m ³
Operations	Lab/site thinking	Continuous ascent, g-load, restart, airport licensing

Gap in one line: CHARM has a credible **aneutronic architecture story**; this vehicle needs a **closed, flight-packaged, gigawatt, high specific-power product** that does not yet exist on paper as an engineered BOM.

13.3 Unobtainiums in the gap (pointer)

Call a requirement **unobtainium** if it is not implied by present CHARM results and would break the SSTO close if false. Two disjoint classes exist:

- **Physics/architecture unobtainiums** — integrated p -¹¹B power balance with chambered species/ash removal/tolerable radiation losses simultaneously, centrifugal differential confinement at useful density and confinement time, selective RF/ponderomotive walls at acceptable recirculating power, wave-mediated ash extraction fast enough to avoid helium poisoning, and ultra-high-DC/open-field electrode structures under continuous α /X-ray load — shared with any CHARM plant regardless of vehicle. Itemized in full, with citations, in the companion reactor paper [16] §13, alongside the flight-weight-cryocooler and RF/backbone-structure packaging unobtainiums that follow directly from the §9.3 mass roll-up.
- **Aerospace packaging / propulsion unobtainiums** — continuous GW-class operation through a multi-hour ascent, pilot-string light-off/space restart at 50–200 kWh class, and the Stage-2/3 thruster packaging ($\sim$ 200–300 kW/kg) and Stage-3 I_{sp}/η_{jet} combination — imposed by the SSTO mission rather than by CHARM itself. Itemized in full, with citations, in the companion engine paper [17] §13.

If the physics/architecture unobtainiums fail, no amount of airframe cleverness saves this mission; if they hold but the packaging unobtainiums fail, CHARM may still be a viable ground plant while this spaceplane does not close.

13.4 How to read the rest of this paper

Sections 1–12 are a **requirements mirror** held up to CHARM: they say what a successful bottle must look like to fly Shuttle-class cargo from a municipal runway to ISS altitude. They are **not** a claim that Pale Blue / Princeton has those numbers. Closing the gap is future plasma physics, materials, and packaging work—tracked against the unobtainium list above.

14 Conclusion

Design goals fix a **Shuttle-style SSTO** with a **real cargo bay**, municipal runway ops, and ISS-altitude LEO. Closing $P_\star = 1$ GW at $\alpha_C = 15$ kW/kg yields a reference vehicle of about 196 t **dry**, 44 t **water**, 240 t **GLOW**. Vision figures (Figs. ~3–10) show the interior floorplan, exterior profile, drop-in cutaways, and assembly trees that those numbers must fit.

The plant that closes the mass/energy budget must deliver, as a single flight-packaged island: continuous p -¹¹B fusion with multi-chamber rotating-mirror confinement onto a 1 GW

electrical bus, ~ 5 GWh-class mission source energy per ascent, an island mass/volume implying $\alpha_C \geq 15$ kW/kg and $\bar{p}_C \gtrsim 8$ MW/m³, light-off/space restart at 50–500 kWh class, and continuous throttleable duty through climb and insertion (§9). The engine that closes the propulsion side must deliver three stage powers around 1 GW peak inside a 15 t mass hole, with stage-2/3 packaging specific power (~ 230 – 320 kW/kg) well beyond demonstrated hardware (§10). These are requirements mirrors, not claims that present CHARM or propulsion hardware already has them — the honest gap, itemized with citations, is carried by the two companion papers: **“Compact CHARM p-¹¹B fusion reactor for electricity generation for aerospace propulsion”** [16] for the reactor unobtainiums (flight-weight cryogenics, REBCO/cryostat structure under ascent loads, DEC electrode structures, the unsized RF/backbone-structure remainder) and **“Three cycle electric SSTO rocket engine”** [17] for the propulsion unobtainiums (stage-1 ducted-fan packaging is comparatively vanilla engineering; stages 2–3 microwave-air-plasma and water-plasma packaging are the open items).

In short: if CHARM physics and reactor packaging close (per [16]) but engine stage-2/3 packaging fails (per [17]), CHARM may still be a viable ground plant while this spaceplane does not close — and vice versa. This vehicle paper’s contribution is the mass/energy closure that ties the two together (§4–§8, §11–§12), not a claim that either companion technology is already in hand.

Acknowledgments

CHARM denotes the chambered aneutronic rotating-mirror architecture developed at Princeton Plasma Physics Laboratory (PPPL) under the ARPA-E economical p -¹¹B program [1]–[11] and discussed toward Pale Blue Fusion. This vehicle sketch is an independent systems exercise and does not speak for that program.

14.1 AI-assisted development

The human author retains sole responsibility for the vehicle-sizing methodology, every modeling choice, and every technical claim in this work; no large language model or image model is listed as a co-author. We gratefully acknowledge assistance from the following tools (auto-generated from `scripts/ai_model_cards.py` when building `arxiv.tex`):

- **Cursor** [18] — agent-assisted development environment for this repository: the OpenVSP floorplan/profile CAD and Blender drop-in-cutaway pipeline, the `constants_model.py` mass/energy-closure derivations, the interactive assembly outliner, and drafting/maintaining this narrative (`arxiv.md`), including splitting the original monorepo into this vehicle-integration paper plus the companion CHARM reactor and combined-cycle-engine papers.
- **xAI Grok 4.5** [19] — primary agent model for large stretches of the vehicle-packaging work: OpenVSP geometry iteration (body flap, OMS pods), the CHARM/engine bottom-up mass roll-up in `constants_model.py`, and the `assembly.json`-driven Blender drop-in figures. Used via the Cursor agent environment.
- **Anthropic Claude Sonnet 5** [20] — session work including narrative maintenance (table of contents, appendix reordering), rewriting §9/§10 into short vehicle-level summaries that cite the companion reactor and engine papers for full derivations, pruning the generic multi-architecture survey scaffold from this repository, and this AI-assistance disclosure. Used via the Cursor agent environment.

- **Google Nano Banana Pro (Gemini 3 Pro Image)** [21] — generated Fig.~1, an illustrative exterior concept render of the vehicle taking off from a municipal airport, and Fig.~2, a companion flight-deck interior render at the same moment, both prompted from the CAD floorplan/profile wireframes as reference images (Figs.~3, 4). Concept art only — proportions, panel lines, cabin layout, and the runway scene are AI interpretation, not CAD; the wireframe figures remain the dimensioned source of truth.

All derived numbers, pipeline code, and final prose were reviewed by the human author, who takes full responsibility for them.

References

- [1] N. J. Fisch et al. (Princeton Plasma Physics Laboratory), “Why pB11?” ARPA-E Fusion Annual Meeting slides (Day2_08_Fisch.pdf), Aug. 2025. Primary public overview of the Chambered Aneutronic Rotating Mirror (CHARM) / chambered rotating-mirror p - ^{11}B architecture. [Online]. Available: https://arpa-e.energy.gov/sites/default/files/2025-08/Day2_08_Fisch.pdf
- [2] Advanced Research Projects Agency–Energy (ARPA-E), “Economical Proton-Boron11 Fusion,” Award No. DE-AR0001554, OPEN 2021.
- [3] E. J. Kolmes, I. E. Ochs, and N. J. Fisch, “Wave-supported hybrid fast-thermal p - ^{11}B fusion,” *Phys. Plasmas*, vol. 29, no. 11, Art. no. 110701, 2022, doi: 10.1063/5.0118337.
- [4] I. E. Ochs, E. J. Kolmes, M. E. Mlodik, T. Rubin, and N. J. Fisch, “Improving the feasibility of economical proton–boron-11 fusion via alpha channeling with a hybrid fast and thermal proton scheme,” arXiv:2210.08076 [physics.plasm-ph], 2022.
- [5] I. E. Ochs and N. J. Fisch, “Lowering the reactor breakeven requirements for proton–boron 11 fusion,” *Phys. Plasmas*, 2024, doi: 10.1063/5.0184945. (ARPA-E DE-AR0001554.)
- [6] I. E. Ochs, E. J. Kolmes, and N. J. Fisch, “Preventing ash from poisoning proton–boron 11 fusion,” *Phys. Plasmas*, 2025. [Online]. Available: <https://w3.pppl.gov/~fisch/fischpapers/2025/Ochs.poisoning>
- [7] I. E. Ochs, E. J. Kolmes, and N. J. Fisch, “On the feasibility of radiation-trapping regimes in compressed proton-boron-11 plasmas,” *Phys. Plasmas*, vol. 32, no. 2, Art. no. 022504, 2025, doi: 10.1063/5.024504.
- [8] N. J. Fisch, I. E. Ochs, E. J. Kolmes, M. E. Mlodik, and T. Rubin, “Nonthermal Proton-Boron11 Fusion with Separated Reactant Regions,” U.S. Patent Application 19/083,790, filed Mar. 19, 2025.
- [9] T. Rubin, J.-M. Rax, N. J. Fisch, I. E. Ochs, and E. J. Kolmes, “Enhanced Particle Confinement with Positive and Negative Ponderomotive Potentials,” U.S. Patent Application 19/084,168, filed Mar. 19, 2025.
- [10] N. J. Fisch et al., “Systems and Methods for Producing Ultra-high DC Voltages in Open Field Line Traps...,” U.S. Patent Application 19/175,473, filed Apr. 10, 2025.
- [11] E. J. Kolmes, I. E. Ochs, and N. J. Fisch, “Method and Apparatus for Differential Confinement, Mixing, and Demixing...,” U.S. Provisional Patent 63/794,470, filed Apr. 25, 2025.
- [12] National Aeronautics and Space Administration, *Space Shuttle Orbiter / OV-105 Atlantis* dimensional and mass data (cargo bay 18.3×4.6 m; payload to LEO $\approx 24\,400$ kg; empty mass $\approx 78\,000$ kg class). NASA fact sheets and Orbiter inventory documentation.
- [13] NASA, *Space Shuttle Vehicle Familiarization* (SSV FAM), crew module description: flight deck; middeck galley, personal hygiene, airlock, and **side hatch** for ground ingress/egress; equipment bay ECLSS. Training document SSV-FAM-1107 and NASA *Space Shuttle News Reference* crew-cabin arrangement figures (flight deck p. 3-9; middeck p. 3-10).

[14] D. Ye, J. Li, and J. Tang, “Jet propulsion by microwave air plasma in the atmosphere,” *AIP Advances*, vol. 10, no. 5, Art. no. 055002, 2020, doi: 10.1063/5.0005814. (Stage-2 anchor: magnetron $\rightarrow$ compressed-air microwave plasma jet.)

[15] Y. Nakagawa, H. Koizumi, H. Kawahara, and K. Komurasaki, “Performance characterization of a miniature microwave discharge ion thruster operated with water,” *Acta Astronautica*, vol. 157, pp. 294–299, 2019, doi: 10.1016/j.actaastro.2018.12.031. (Stage-3 architecture: water + microwave plasma; demo $I_{sp} \sim 665$ s, μ N thrust.)

[16] L. W. Ericson, “Compact CHARM p-¹¹B fusion reactor for electricity generation for aerospace propulsion,” Catskills Research Company, 2026. Companion reactor paper: bottom-up magnet/cryo/shield mass roll-up and reactor-side unobtainiums. [Online]. Available: https://github.com/catskillsresearch/compact_p11b

[17] L. W. Ericson, “Three cycle electric SSTO rocket engine,” Catskills Research Company, 2026. Companion engine paper: closed-form stage 1/2/3 power/mass/duration solve, aerodynamics/CFD cross-check, ascent trajectory, acoustic signature, and propulsion-packaging unobtainiums. [Online]. Available: https://github.com/catskillsresearch/electric_3_stage_ssto_engine

AI development-tool references (§ Acknowledgments, auto-generated from `scripts/ai_model_cards.py`):

[18] Anysphere, Inc. *Cursor: AI-native code editor and agent environment*. <https://cursor.com> (accessed 2026).

[19] xAI. *Grok 4.5*. Model documentation, <https://docs.x.ai/developers/models/grok-4.5>; Cursor announcement, <https://cursor.com/blog/grok-4-5> (accessed 2026).

[20] Anthropic. *Claude Sonnet 5*. System card, <https://www.anthropic.com/claude-sonnet-5-system-card>; model documentation as integrated in Cursor, <https://cursor.com/docs/models> (accessed 2026).

[21] Google DeepMind. *Nano Banana Pro (Gemini 3 Pro Image)*. Product announcement, <https://blog.google/innovation-and-ai/products/nano-banana-pro/>; model documentation, <https://ai.google.dev/gemini-api/docs/image-generation> (accessed 2026).

A Design software

This appendix inventories **software still in active use** for the vehicle packaging model and this paper’s figure/PDF pipeline. Tools tried and discarded are omitted. Further subsections will be added as the toolchain grows.

A.1 Imported packages actually used

Table 17 External packages and tools used by the living design / paper build.

Package / tool	Role in this project
Python $\geq$ 3.12 + Poetry	Project environment; CAD scripts; assembly JSON tooling; paper build driver
NumPy	OpenVSP figure export; <code>constants_model.py</code> sizing
Matplotlib	OpenVSP floorplan/profile rasters
OpenVSP (optional Poetry group; upstream <code>.deb</code> + Python API; NOSA 1.3)	Parametric vehicle CAD (<code>.vsp3</code>); source of the orthographic floorplan and profile figures

Package / tool	Role in this project
Blender 5.x (snap /snap/bin/blender)	Drop-in cutaways from <code>assembly.json</code> (crew capsule, airlock, cargo skid, fusion plant skid top-down; <code>make cad-drop-ins</code>)
NumPy (research/figures/cad/constants_model.py)	Single source for every sizing-constraint number in §6–§9 and the CHARM bottom-up mass roll-up (§9.3); regenerates ‘spans in this file and patchesassembly.json/vehicle_spec.json- plain arithmetic, never an LLM call **Pillow** Image handling when the paper build ingests raster figure assets **Pandoc** arxiv.md→ LaTeX body conversion insidescripts/build_arxiv_tex.py **Mermaid CLI** (mmdc/@mermaid-js/mermaid-cli) Paper mermaid fences →figures/figure-NNN.pdf **latexmk** + **LuaLaTeX** Localarxiv.pdf/zenodo.pdfcompile (.latexmkrc‘)
Mermaid.js v11 (CDN, browser)	Live diagram engine inside the assembly outliner (§A.2)

Python’s standard library (**json**, **http.server**, etc.) serves assembly I/O and the outliner static server; it is not listed as an imported third-party package.

The VSPAERO/OpenFOAM/SU2 aerodynamics-analysis toolchain (Stage-1 outline check, digital-wind-tunnel polar, CFD meshing) is **not** part of this repository — it lives in the companion engine paper [17], which reproduces those figures from its own copy of `constants_model.py`.

A.2 Assembly outliner

The **assembly outliner** is a small local web app under `research/figures/cad/hierarchy_app/`. It is the interactive view of the vehicle packaging tree.

Source of truth. `research/figures/cad/assembly.json` holds the hierarchy (collections vs parts), ports, and joints. Paper mermaid plant/station figures are reconciled to this file (§1). A companion emitter writes `assembly_hierarchy.mmd` for offline diffs.

UI. Left pane: Blender-style tree (expand/collapse, collections marked). Right pane: Mermaid flowchart of the **visible** subtree — grey containment edges, teal functional joints; collapsed mates **proxy** to the nearest expanded ancestor; neighboring expanded siblings get distinct tints so stage-1/2/3 (and plant sub-racks) read at a glance.

How to run. From the repo root: `make cad-outliner` (or `./research/figures/cad/serve_hierarchy_app.sh`) → open `http://127.0.0.1:8765/hierarchy_app/`. The server sends `Cache-Control: no-store`; use **Reload data** after editing `assembly.json`.

Stack. Static HTML/CSS/JS; Mermaid.js in the browser; no build step. The serve script is a tiny Python `ThreadingHTTPServer`.

A.3 Blender drop-in cutaways

Layout-critical packaging figures (hatches, seat rows, aisle clearances, bay doors, magnet/cryo layout) are built as **editable Blender geometry** from `assembly.json`, not AI image prompts. Four modules are done this way — crew capsule, airlock, cargo skid, and fusion plant skid:

```
make cad-drop-ins
# research/figures/{crew_capsule_top,airlock_top,cargo_skid_top,fusion_plant_skid_top}.png
# research/figures/cad/{crew_capsule_cutaway,airlock_cutaway,cargo_skid_cutaway,
  fusion_plant_skid_cutaway}.blend
```

```
./bl.sh # GUI edit (crew capsule; edit the path for the others)
```

A.4 Generated numeric constants

Every numeric value wrapped in a machine-readable HTML comment pair in this document (the CHARM mass/power/cryo chain of §6–§9, most visibly §9.6) is **program-controlled**, not hand-typed: it is written by `scripts/update_arxiv_constants.py`, which re-runs `research/figures/cad/constants_model.py` (pure NumPy + Python stdlib, no LLM in the loop) and regex-replaces only the text between each marker pair, leaving surrounding prose untouched. The same run also writes `research/figures/cad/constants.generated.json`, which `scripts/apply_constants_to_assembly.py` reads to patch `assembly.json`’s magnet/cryocooler node counts and `size` blocks, and which `build_fusion_plant_skid_blender.py` reads for its magnet/cryocooler counts — so the paper, the JSON single source of truth, and the Blender renders can never numerically disagree. `make paper-render` runs the whole chain; it is a dependency of `make arxiv` and `make zenodo-tex`.

Each figure has its own placement script — `build_crew_capsule_blender.py`, `build_airlock_blender.py`, `build_cargo_skid_blender.py` — sharing common primitives, hatch/shell/door “kits”, and camera setup from `research/figures/cad/lib/` (`assembly_parser.py`, `procedural_geometry.py`, `render_utils.py`). The fusion-plant skid (CHARM chambers) and combined-cycle engine skid (scoops/duct/nozzle/water tanks) are not yet migrated — those need real procedural modeling of hardware that has no library equivalent, and remain AI concept sketches for now.

Working draft for HAL.science deposit. Not a license to operate a reactor.